

LABTASK 1:

SOURCE CODE:

```
#include <iostream>
using namespace std;

✓ class paymentSystem{
protected:
    double amount;
    double serviceFee;
    double totalAmount;
    double refundAmount;
public:
    virtual void pay()=0;
    virtual void refund()=0;

    ✓ void set_amount(){
        cout<<"ENTER AMOUNT: ";
        cin>>amount;
    }
};
```

```
class EasyPaize: public paymentSystem{
public:

    void pay() override{
        cout<<"EASYPAISA"<<endl;
        serviceFee = amount * 0.02;
        totalAmount = amount + serviceFee;
        cout<<"ORIGINAL AMOUNT: "<<amount<<endl;
        cout<<"SERVICE FEE: "<<serviceFee<<endl;
        cout<<"FINAL AMOUNT: "<<totalAmount<<endl;
    }
    void refund() override {
        serviceFee = amount * 0.02;
        refundAmount = amount - serviceFee;
        cout<<"REFUND AMOUNT: "<<refundAmount<<endl;
    }
};
```

```
class JazzCash: public paymentSystem{

public:
    void pay() override{
        cout<<"JAZZCASH"<<endl;
        serviceFee = amount * 0.05;
        totalAmount = amount + serviceFee;
        cout<<"ORIGINAL AMOUNT: "<<amount<<endl;
        cout<<"SERVICE FEE: "<<serviceFee<<endl;
        cout<<"FINAL AMOUNT: "<<totalAmount<<endl;
    }
    void refund() override
    {
        serviceFee = amount * 0.05;
        refundAmount = amount - serviceFee;
        cout<<"REFUND AMOUNT: "<<refundAmount<<endl;
    }
};
```

```
int main(){
    paymentSystem *ps;
    EasyPaise eobj;
    JazzCash jobj;
    char paymentsys;
    char select_again;
    do
    {
        cout<<"SELECT YOUR PAYMENT SYSTEM"<<endl;
        cin>>paymentsys;

        if(paymentsys == 'e')
        {
            ps = &eobj;
            ps->set_amount();
            ps->pay();
            ps->refund();
        }
        else if(paymentsys=='j')
```

```

        ps->refund();
    }
    else if(paymentsys=='j')
    {
        ps = &jobj;
        ps->set_amount();
        ps->pay();
        ps->refund();
    }
    else
    {
        cout<<"INVALID OPTION - SELECT AGAIN";
    }
    cout<<"WANT ANOTHER TRANSACTION? (y/n): "<<endl;
    cin>>select_again;
} while(select_again=='y');

return 0;
}

```

OUTPUT:

```

SELECT YOUR PAYMENT SYSTEM
j
ENTER AMOUNT: 50000
JAZZCASH
ORIGINAL AMOUNT: 50000
SERVICE FEE: 2500
FINAL AMOUNT: 52500
REFUND AMOUNT: 47500
WANT ANOTHER TRANSACTION? (y/n):
y

```

```
5
SELECT YOUR PAYMENT SYSTEM
e
ENTER AMOUNT: 1500
EASYPAlSA
ORIGINAL AMOUNT: 1500
SERVICE FEE: 30
FINAL AMOUNT: 1530
REFUND AMOUNT: 1470
WANT ANOTHER TRANSACTION? (y/n):
n
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```

LABTASK 2:

```
#include <iostream>
using namespace std;

class FeeSystem{
protected:
    double amount;
    double processingCharge;
    double final_amount;
    int extraCharges;
public:
    virtual void submitFee()= 0;
    virtual void returnFee()= 0;
};
```

```
class BankTransfer: public FeeSystem{
public:

    void submitFee() override{
        cout<<"ENTER FEE TO SUBMIT: "<<endl;
        cin>>amount;
        extraCharges=100;
        final_amount = amount + extraCharges;
        cout<<"FEE AMOUNT: "<<amount<<endl;
        cout<<"EXTRA CHARGES: "<<extraCharges<<endl;
        cout<<"TOTAL AMOUNT PAID: "<<final_amount<<endl;
    }

    void returnFee() override{
        cout<<"REFUND FEE: "<<amount<<endl;
    }
};
```

```
class CreditCard: public FeeSystem{  
    void submitFee() override{  
        cout<<"ENTER FEE TO SUBMIT: "<<endl;  
        cin>>amount;  
        processingCharge= amount*0.03;  
        final_amount = amount + processingCharge;  
        cout<<"FEE AMOUNT: "<<amount<<endl;  
        cout<<"EXTRA CHARGES: "<<processingCharge<<endl;  
        cout<<"TOTAL AMOUNT PAID: "<<final_amount<<endl;  
    }  
  
    void returnFee() override{  
        cout<<"REFUND FEE: "<<amount<<endl;  
    }  
};
```

```

int main(){
    FeeSystem *fs;
    BankTransfer bobj;
    CreditCard cobj;
    char paymentMethod;
    char select_again;
    do
    {
        cout<<"SELECT YOUR PAYMENT METHOD"<<endl;
        cin>>paymentMethod;

        if(paymentMethod == 'b')
        {
            fs = &bobj;
            fs->submitFee();
            fs->returnFee();
        }
        else if(paymentMethod=='c')
        {
            fs = &cobj;
            fs->submitFee();
            fs->returnFee();
        }
    }

```

```

    }
    else
    {
        cout<<"INVALID OPTION - SELECT AGAIN";
    }
    cout<<"WANT ANOTHER TRANSACTION? (y/n): "<<endl;
    cin>>select_again;
} while(select_again=='y');

return 0;
}

```


OUTPUT:

```
SELECT YOUR PAYMENT METHOD
b
ENTER FEE TO SUBMIT:
10000
FEE AMOUNT: 10000
EXTRA CHARGES: 100
TOTAL AMOUNT PAID: 10100
REFUND FEE: 10000
WANT ANOTHER TRANSACTION? (y/n):
y
```

```
SELECT YOUR PAYMENT METHOD
c
ENTER FEE TO SUBMIT:
22000
FEE AMOUNT: 22000
EXTRA CHARGES: 660
TOTAL AMOUNT PAID: 22660
REFUND FEE: 22000
WANT ANOTHER TRANSACTION? (y/n):
n
```